DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

Jaxin Lu, Haotian Pu, Julia Lee
Github: https://github.com/j363lu/dreambooth

1. Introduction

In this project, we attempt to reproduce the result from ***DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation*** (Ruiz et al., 2023). Diffusion models can generate images given a text prompt. However, they cannot generate specific subjects (e.g. my own dog). The paper proposed a way to fine-tune the models on a few images (roughly 5) of a subject to enable the generation of the subject.

Figure 1: Overview of results from our fine-tuned model

2. Chosen Result

We intend to reproduce the fine-tuning pipeline in the paper, fine-tune a few models to generate some visual results, and implement the metrics in Table 1 below. Since Imagen is a closed-source model, we only fine-tune on stable diffusion models.

Method	DINO ↑	CLIP-I ↑	CLIP-T ↑
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	**0.696**	**0.812**	**0.306**
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255

Table 1: Results of the original paper

3. Methodology

3.1 Dataset

We use the dataset provided by the original paper, which contains 30 different subjects, with roughly 5 images for each subject. We also collected images for one new subject ourselves with 5 images.

3.2 Training objective

We follow the same methodology described in the original paper to fine-tune diffusion models. There are two losses: (1) a reconstruction loss, and (2) a class-specific prior preservation loss.

The reconstruction loss encourages the model to generate images of the same subject as the given subject images. A special token “sks” is included in the prompt to refer to the specific subject. For example, when diffusion model is prompted “a sks dog”, it should generate images for the specific subject dog. The 5 images for the subject are used to supervise the model output when prompted “a sks dog”.

The prior preservation loss ensures the model does not overfit on the given subject images. We generate 100 general images with the same class as the subject using the diffusion model before fine-tuning (e.g. 100 general dog images using prompt “a dog”). These images are used to supervise the model with the prompt “a dog”, to preserve the model’s prior on dogs. This allows the model to still generate general dogs after finetuning.

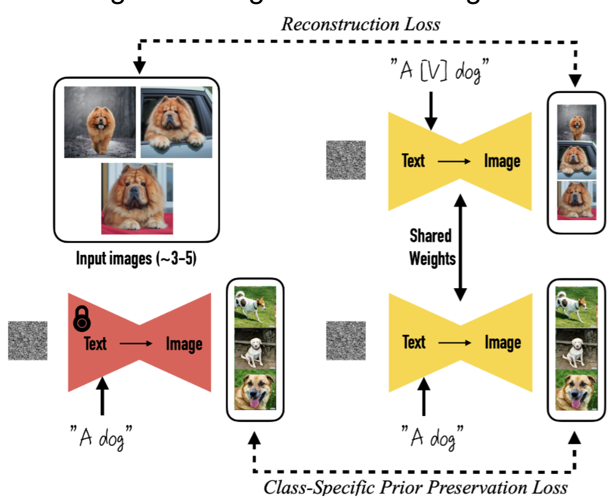

Figure 2: Training objective

3.3 Evaluation

Subject fidelity and prompt fidelity are used to evaluate the result. Subject fidelity evaluates whether the subject generated by the model is the same as the original 5 images of the subject. We use DINO and CLIP to encode the original images and the generated images and calculate the similarity. If the similarity is high, the generated images contain the same subject as the original.

Prompt fidelity measures how well the generated images follow the prompt. We use CLIP to encode the generated images and the prompt and calculate the similarity. If the similarity is high, the model follows the instruction well.

4. Results & Analysis

4.1 Qualitative Results

The following figures show the provided subject images and generated results from fine-tuned models. A different stable-diffusion-v1.5 instance was fine-tuned for each subject. We experimented modifying the expression, viewpoint, background, and outfit of the subjects.

Expression Modification (“A [state] sks dog”)

Figure 3: Expression modification

Viewpoint Modification (“A sks cat seen from the [viewpoint]”)

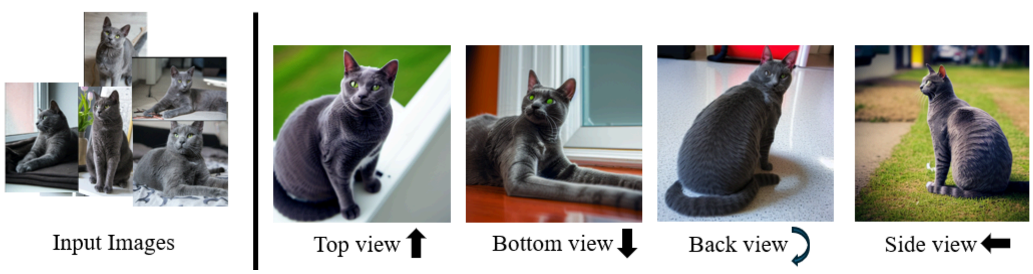

Figure 4: Viewpoint modification

Background Modification (“A sks bottle [background]”)

Figure 5: Background modification

Figure 6: Outfit modification

We experimented fine-tuning models with and without prior preservation loss. Without prior preservation loss, the fine-tuned model can no longer generate a general robot toy when prompted to generate "a robot toy". With prior preservation loss, the model can still generate general robot toys.

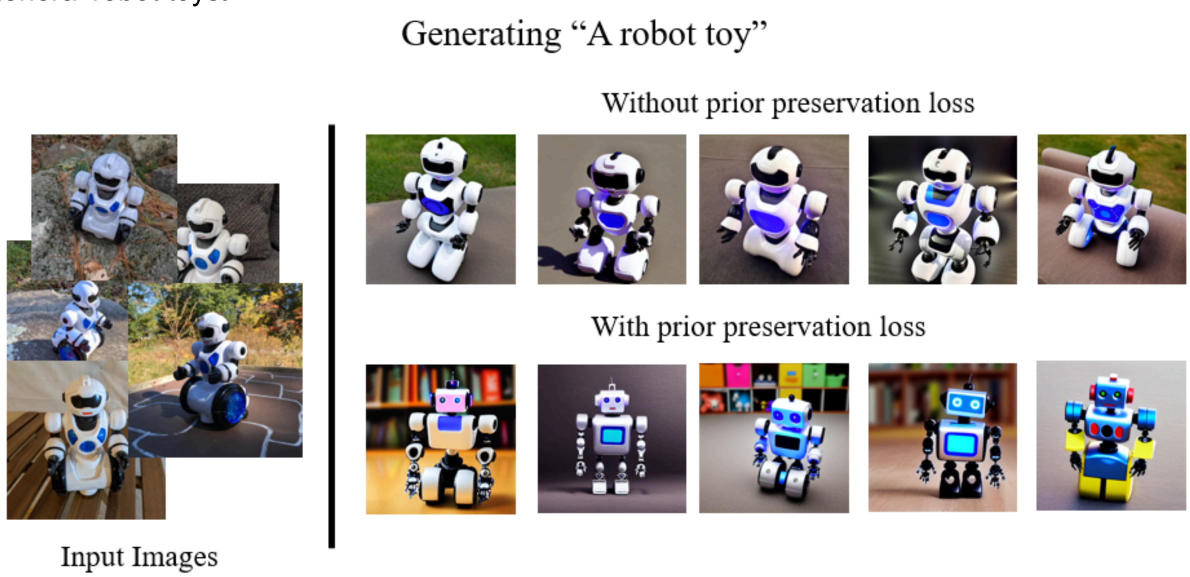

Figure 7: With and without prior preservation loss

4.2 Quantitative results

Table 1 shows subject fidelity using DINO and prompt fidelity using CLIP

Method / Category	DINO	CLIP-I	CLIP-T
Reported paper baselines			
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255
Local reproduction results			
bottle	**0.799**	0.817	0.269
cat2	0.768	**0.890**	0.276
dog8	0.666	0.885	0.273
duck_toy	0.714	0.862	**0.326**
robot_toy	0.733	0.842	0.285
Average	0.736	0.859	0.286

Table 2: Quantitative results

Compared to the DreamBooth paper baselines, the reproduction shows better DINO and CLIP-I performance but lower CLIP-T performance on average. The results support the core DreamBooth claim that subject-driven fine-tuning can produce personalized outputs with good visual fidelity to the reference images.

The main challenge during reimplementation is hyperparameter tuning. The original paper does not disclose implementation details, so we had to tune the parameters ourselves.

5. Reflections

The key takeaway from the project is that diffusion models can be fine-tuned to generate a specific subject with only a few examples. Potential future extensions include using LoRA to fine-tune diffusion models to reduce cost and fine-tuning larger and newer diffusion models than stable-diffusion-v1.5.

6. References

Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., & Aberman, K. (2023). *DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 22500-22510).